Walchand Institute of Technology, Solapur
(An Autonomous Institute)

Affiliated to
Punyashlok Ahilyadevi Holkar Solapur University,
Solapur

Choice Based Credit System (CBCS)

Structure and Syllabus
for
Final Year B.Tech.
Electronics and Telecommunication Engineering
for
Academic Year 2026-27

HEAD
Electronics & Telecommunication Department
Walchand Institute of Technology
SOLAPUR - 413006

Dean (Academics)

WIT SOLAPUR (Autonomous) WALCHAND INSTITUTE OF TECHNOLOGY

Department of Electronics and Telecommunication Engineering

Vision

- To be a distinguished center for nurturing the holistic development of competent young engineers in the electronics and allied field.

Mission

- **M1:** To inculcate and stimulate Electronics & allied Engineering proficiency amongst students through quality education and innovative educational practices.
- **M2:** To create engineering professionals with social consciousness.
- **M3:** To foster technical skills of students through creativity and critical thinking.
- **M4:** To enhance soft skill set of students which is crucial for career success through effectual training.

Department of Electronics and Telecommunication Engineering

Program Educational Objectives (PEOs)

- Graduates will exhibit strong fundamental knowledge and technical skills in Electronics and Telecommunication Engineering and allied fields.
- Graduates will manifest technological progression, hardware & software skills to fabricate sustainable, energy efficient and futuristic solutions to pursue successful professional careers in multidisciplinary fields.
- Graduates will demonstrate professional ethics, effective communication, teamwork, leadership qualities and ability to relate engineering issues to broader social context along with lifelong learning.

Knowledge and Attitude Profile (WK)

WK1	A systematic, theory-based understanding of the natural sciences applicable to the discipline and awareness of relevant social sciences.
WK2	Conceptually-based mathematics, numerical analysis, data analysis, statistics and formal aspects of computer and information science to support detailed analysis and modelling applicable to the discipline.
WK3	A systematic, theory-based formulation of engineering fundamentals required in the engineering discipline.
WK4	Engineering specialist knowledge that provides theoretical frameworks and bodies of knowledge for the accepted practice areas in the engineering discipline; much is at the forefront of the discipline.
WK5	Knowledge, including efficient resource use, environmental impacts, whole-life cost, re-use of resources, net zero carbon, and similar concepts, that supports engineering design and operations in a practice area.
WK6	Knowledge of engineering practice (technology) in the practice areas in the engineering discipline.
WK7	Knowledge of the role of engineering in society and identified issues in engineering practice in the discipline, such as the professional responsibility of an engineer to public safety and sustainable development.
WK8	Engagement with selected knowledge in the current research literature of the discipline, awareness of the power of critical thinking and creative approaches to evaluate emerging issues.
WK9	Ethics, inclusive behavior and conduct. Knowledge of professional ethics, responsibilities, and norms of engineering practice. Awareness of the need for diversity by reason of ethnicity, gender, age, physical ability etc. with mutual understanding and respect, and of inclusive attitudes.

Program Outcomes (POs)	
PO 1	**Engineering Knowledge:** Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization as specified in WK1 to WK4 respectively to develop to the solution of complex engineering problems.
PO 2	**Problem Analysis:** Identify, formulate, review research literature and analyze complex engineering problems reaching substantiated conclusions with consideration for sustainable development. (WK1 to WK4)
PO 3	**Design/Development of Solutions:** Design creative solutions for complex engineering problems and design/develop systems/components/processes to meet identified needs with consideration for the public health and safety, whole-life cost, net zero carbon, culture, society and environment as required. (WK5)
PO 4	**Conduct Investigations of Complex Problems:** Conduct investigations of complex engineering problems using research-based knowledge including design of experiments, modelling, analysis & interpretation of data to provide valid conclusions. (WK8).
PO 5	**Engineering Tool Usage:** Create, select and apply appropriate techniques, resources and modern engineering & IT tools, including prediction and modelling recognizing their limitations to solve complex engineering problems. (WK2 and WK6)
PO 6	**The Engineer and The World:** Analyze and evaluate societal and environmental aspects while solving complex engineering problems for its impact on sustainability with reference to economy, health, safety, legal framework, culture and environment. (WK1, WK5, and WK7).
PO 7	**Ethics:** Apply ethical principles and commit to professional ethics, human values, diversity and inclusion; adhere to national & international laws. (WK9)
PO 8	**Individual and Collaborative Team work:** Function effectively as an individual, and as a member or leader in diverse/multi-disciplinary teams.
PO 9	**Communication:** Communicate effectively and inclusively within the engineering community and society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations considering cultural, language, and learning difference
PO 10	**Project Management and Finance:** Apply knowledge and understanding of engineering management principles and economic decision-making and apply these to one’s own work, as a member and leader in a team, and to manage projects and in multidisciplinary environments.
PO 11	**Life-long Learning:** Recognize the need for, and have the preparation and ability for i) independent and life-long learning ii) adaptability to new and

	emerging technologies and iii) critical thinking in the broadest context of technological change. (WK8)

Program Specific Outcomes (PSOs)

- Graduates will be able to attain a solid foundation in Electronics and Telecommunication Engineering with an ability to function in multidisciplinary environment.
- Graduates will be able to use techniques and skills to design, analyze, synthesize, and simulate Electronics and Telecommunication Engineering components and systems.
- Graduate will be capable of developing programs in Assembly, High level and HDL languages using contemporary tools for software development.

Department of Electronics and Telecommunication Engineering	
Legends Used	
L	Lecture Hours / week
T	Tutorial Hours / week
P	Practical Hours / week
FA	Formative Assessment
SA	Summative Assessment
ESE	End Semester Examination
ISE	In Semester Evaluation
ICA	Internal Continuous Assessment
POE	Practical and Oral Exam
OE	Oral Exam
MOOC	Massive Open Online Course
HSS	Humanity and Social Science
NPTEL	National Programme on Technology Enhanced Learning
F.Y.	First Year
S.Y.	Second Year
T.Y.	Third Year
B.Tech.	Bachelor of Technology

Department of
Electronics and Telecommunication Engineering

Course Code Format

2	3	E	T	U/P	2	C	C	1	T/L
Year of Admission		Program Code		U-Under Graduate, P-Post Graduate	Semester No./ Year1/2/3/…8	Course Type		Course Serial No. 1-9	T-Theory, L-Lab session A-Tutorial P-Programming / Design

Program Code	
ET	Electronics and Telecommunication Engineering
Course Type	
BS	Basic Science
ES	Engineering Science
HU	Humanities & Social Science
MC	Mandatory Course
CC	Programme Core Compulsory Course
SN*	Self-Learning (**N*** *indicates the serial number of electives offered in the respective category)*
EN*	Programme Core Elective Course (**N*** *indicates the serial number of electives offered in the respective category)*
SK	Skill-Based Course
SM	Seminar
MP	Mini project
PR	Project
IN	Internship
ON*	Open Elective (**N*** *indicates the serial number of electives offered in the respective category)*
MD	Multidisciplinary Minor
EM	Entrepreneurship/Economics/Management Courses
FP	Community Engagement Project / Field Project
AE	Ability Enhancement Course
VE	Value Education Course
IK	Indian Knowledge System
VS	Vocational & Skill Enhancement Course

RM	Research Methodology
HN	Honors' Degree Course
HR	Honors Research

Sample Course Code	
23ETU5CC1T	CMOS Technology

Department of Electronics and Telecommunication Engineering

B. Tech. Semester VII 2026-27

Course Code	Name of Course	Engagement Hours			Credits	SA		FA		Total
		L	T	P		Theory	OE/ POE	ISE	ICA	
23ETU7CC1T	CMOS Technology	3		-	3	60		40		100
23ETU7EN*2T	Program Elective IV	2		-	2	60		40		100
23ETU7EN*3T	Program Elective V	2		-	2	60		40		100
23ETU7EN*3A	Program Elective V (Tutorial)		1		1				25	25
23ETU7CC4T	Programmable Logic Control	3			3	60		40		100
23##U7MD6T	Multi Disciplinary Minor V	3		-	3	60		40	-	100
23##U7MD6A	Multi Disciplinary Minor V (Tutorial)		1		1				25	25
	Subtotal	**13**	**2**	**-**	**15**	**300**		**200**	**50**	**550**
Laboratory Courses										
23ETU7CC1L	CMOS Technology Lab	-	-	2	1		**25**		25	50
23ETU7EN*2L	Program Elective IV Lab	-	-	2	1				25	25
23ETU7CC4L	Programmable Logic Control Lab			2	1				25	25
23ETU7PR5L	Project Phase I	-	-	8	4		**50**		100	150
	Subtotal		**-**	**14**	**7**		**75**		**175**	**250**
	Grand Total	**13**	**2**	**14**	**22**	**375**		**200**	**225**	**800**

Department of
Electronics and Telecommunication Engineering

B. Tech. Semester VIII 2026-27

Course Code	Name of Course	Engagement Hours			Credits	SA		FA		Total
		L	T	P		Theory	OE/ POE	ISE	ICA	
23CMU8RN*1T	Research Methodology And Intellectual Property Rights / MOOC	-	-	-	2	50	-	-	-	50
23ETU8EN*2T	Program Elective Course VI / MOOC#	-	-	-	2	50	-	-	-	50
	Subtotal	**-**		**-**	**4**	**100**	**-**	**-**	**-**	**100**
Laboratory Courses										
23ETU8IN3L	Internship	-	-	20	10	-	50	-	100	150
23ETU8PR4L	Project Phase II			4	2		50		100	150
	Subtotal	**-**	**-**	**24**	**12**	**-**	**100**	**-**	**200**	**300**
	Grand Total	**-**	**-**	**24**	**16**	**200**		**-**	**200**	**400**

Note:

- N* indicates the serial number of electives offered in the respective category
- ## indicates program code of offering Program
- *All lectures will be in online mode.
- # List of Courses will be provided by the BoS.

- **List of Program Elective IV**

List of Program Elective IV (Semester –VII)	
Course Code	**Course title**
23ETU7E12T	Artificial Intelligence
23ETU7E22T	Architecting IoT Solutions
23ETU7E32T	Microwave Engineering

- **List of Program Elective V**

List of Program Elective V (Semester –VII)	
Course Code	**Course Title**
23ETU7E13T	Business Intelligence
23ETU7E23T	Automotive Electronics
23ETU7E33T	Mobile Communication

- **List of Program Elective VI**

List of Program Elective VI (Semester –VIII)	
Course Code	**Course Title**
23ETU8E12T	Artificial Intelligence Applications
23ETU8E22T	Industrial Internet of Things
23ETU8E32T	Programmable ICs and ASIC

- **For Courses offered during Semester VIII**

Sr. No	***Course Code***	***Option -I***	***Option II***
1	23CMU8RN*1T	'Research Methodology & IPR' course offered by the department during Semester VIII in Self Learning / Online mode and successfully passing ESE for the same. 23CMU8R11T – Self Learning Mode	MOOC Certification* 23CMU8R21T – For MOOC (Option I) 23CMU8R31T – For MOOC (Option II)
2	23ETU8EN*2T	'Program Elective VI' course offered by the department during Semester VIII in Self Learning / Online mode and successfully passing ESE for the same. Program Elective VI options: 23ETU8E12T- Artificial Intelligence Applications 23ETU8E22T- Industrial Internet of Things 23ETU8E32T-Programmable ICs and ASIC	MOOC Certification* 23ETU8E42T – MOOC (Option I) 23ETU8E52T – MOOC (Option II) 23ETU8E62T – MOOC (Option III)

* If students have not completed the MOOC in Semester V or VI, then students have to complete this MOOC in the Semester VII or VIII. In Semester VIII, the marks earned by the students in final assessment of this MOOC will be appropriately scaled and transferred for 23CMU8RN*1Tand / or 23ETU8EN*2T along with the required credits. The list of approved MOOCs will be finalized and released by Board of Studies before the commencement of the semester as per the available courses on MOOC platform.

If students do not pass the MOOC examination of 23CMU8RN*1T or 23ETU8EN*2T before commencement of End Semester Examination (ESE) of Semester VIII, then they must appear for the institute-conducted ESE for courses (mentioned in option I) by department during Semester VIII.

- **Multi Disciplinary Minor (MDM) Courses**

Sr. No.	MDM Program	MDM I (Sem III)	MDM II (Sem IV)	MDM III (Sem V)	MDM IV (Sem VI)	MDM V (Sem VII)
1	Computer Science and Engineering	Operating System	UI Technologies	Software Engineering	Big Data Technologies	Software Testing and Quality Assurance
2	Information Technology	Principles of Operating Systems	Web UI and UX Technology	Software Engineering Principles	Data Preprocessing and Visualization	Cyber Security
3	Mechanical and Automation Engineering	Manufacturing Processes and Mechanisms	Machine Drawing and 3D Modeling	Automotive Engineering and Robotics	Additive Manufacturing	Thermal Systems
4	Civil Engineering	Smart Buildings	Geoinformatics	Environmental Impact Assessment	Infrastructural Systems	Disaster Preparedness and planning

23ETU7CC1T: CMOS Technology

Teaching Scheme		Examination Scheme	
Lectures	3 Hours/week	**ESE**	60 Marks
Practical	2 Hours/week	**ISE**	40 Marks
Credits	4	**ICA**	25 Marks
		POE	25 Marks

Introduction:

This course introduces how to design, simulate and test logic circuits using different CMOS Logic Design. It also describes the design of sequential logic circuits and timing issues present in the implementation of the logic circuits applications.

Course Prerequisite:

This course requires knowledge of Digital Devices, combinational and sequential logic circuit design and simulation.

Course Objectives:

1. To make student learn EDA Tools for CMOS Logic Design and simulation.
2. To enable student to design CMOS Logic based design modules for combinational logic circuits.
3. To enable student to design CMOS Logic based design modules for sequential logic circuits.
4. To acquaint students to timing issues, arithmetic and memory module design and testing

Course Outcomes:

At the end of the course, students will be able to

1. Describe MOS transistor theory and behavior of E-MOSFET.
2. Apply design process rules for CMOS circuit layout design.
3. Analyze combinational circuits using CMOS Logic structures.
4. Analyze sequential circuits using CMOS Logic structures and Timing issues in digital circuits.
5. Design arithmetic and memory building blocks using CMOS technology.

Unit – I	MOS Transistor Theory	8 Hours

Physical structure of MOS transistor, accumulation, depletion & inversion modes, MOS device design equations, second order effects, Static and dynamic behavior of CMOS inverter, power and energy delay, Technology scaling, impact of technology scaling on inverter.

Unit – II	Circuit Design Processes	6 Hours

MOS Layers, Stick Diagrams, Design Rules and Layouts – Lambda based design and other rules.

Unit – III	**CMOS Logic Structures for Combinational Logic Design**	**10 Hours**
Static CMOS design- complementary CMOS, Implementation of Boolean Expressions using CMOS Logic, Ratioed logic and pass transistor logic; Dynamic CMOS design- dynamic logic basic principle, speed and power dissipation, issues in dynamic design, cascading dynamic gates, comparison of static and dynamic designs in CMOS.		
Unit – IV	**CMOS Logic Structures for Sequential Logic Design**	**8 Hours**
Static latches and registers- the bistability principle, multiplexer-based latches, Master-slave edge triggered register, low voltage static latches, static SR flip flops, dynamic latches and registers dynamic transmission-gate edge triggered registers, C2MOS- A clock- skew insensitive approach, true single-phase clocked register (TSPCR).		
Unit – V	**Timing Issues in Digital Circuits**	**4 Hours**
Synchronous design- clock skew, jitter, clock distribution, latch-based clocking, synchronizers and arbiters, using PLL for clock synchronization.		
Unit – VI	**Designing Arithmetic and Memory Building Blocks**	**6 Hours**
Designing fast adders, designing fast multipliers, designing other arithmetic building blocks, designing ROMs, DRAMs & SRAMs.		
Internal Continuous Assessment (ICA): ICA consists of minimum of eight experiments based on above syllabus using any EDA software tool suggested.		
Text Books		
1. Digital Integrated Circuits, Rabey, Chandrakasan, Nikolic, Pearson Education 2. CMOS VLSI design, Neil H. E. Weste, David Harris, Ayan Banerjee, Pearson Education.		
Reference Books		
1. CMOS digital integrated circuits, Analysis and Design, Sung-Mo Kang, Yusuf Leblebici, TATA McGRAW Hill 2. Principles of CMOS VLSI Design, Neil Weste, Kamran Eshraghian, Addison Wesley/Pearson Education 3. Modern VLSI Design , Wayne Wolf,, 2nd Edition, Prentice Hall 4. Essentials of VLSI Circuits and Systems – Kamran Ehraghian, Dauglas A. Pucknell and Sholeh Eshraghiam, PHI, EEE		

23ETU7E12T: Program Elective IV - Artificial Intelligence

Teaching Scheme		**Examination Scheme**	
Lectures	2 Hours/week	**ESE**	60 Marks
Practical	2 Hours/week	**ISE**	40 Marks
Credits	3	**ICA**	25 Marks

Introduction:

The Artificial Intelligence course introduces the principles and techniques used to build intelligent systems. It covers AI foundations, problem-solving through search, knowledge representation and reasoning, and learning methods including machine learning and reinforcement learning. The course enables students to understand how rational agents perceive, reason, learn, and make decisions in complex environments.

Course Prerequisite:

This course requires knowledge of discrete mathematics and logic, basic concepts of probability and linear algebra.

Course Objectives:

1. To introduce the fundamental concepts, scope, and foundations of Artificial Intelligence.
2. To develop understanding of problem-solving techniques using search and game-playing strategies.
3. To familiarize students with knowledge representation methods and logical reasoning techniques.
4. To provide knowledge of learning paradigms, machine learning models, and reinforcement learning approaches.

Course Outcomes:

At the end of the course, students will be able to
1. Identify fundamental concepts, foundations, and agent models of Artificial Intelligence.
2. Describe problem-solving approaches and search techniques used in Artificial Intelligence.
3. Illustrate knowledge representation schemes and reasoning techniques for intelligent systems.
4. Outline learning approaches including machine learning and reinforcement learning concepts.

Unit – I	**AI Overview**	**7 Hours**

What Is AI?, The Foundations of Artificial Intelligence, The History of Artificial Intelligence, The State of the Art, Agents and Environments, Good Behavior: The Concept of Rationality, The Nature of Environments, The Structure of Agents

Unit – II	Problem-solving through Search	7 Hours
Problem-Solving Agents, Example Problems, Searching for Solutions, Uninformed Search Strategies, Informed (Heuristic) Search Strategies, Heuristic Functions, Games, Optimal Decisions in Games, Alpha–Beta Pruning, Imperfect Real-Time Decisions, Stochastic Games, Partially Observable Games		
Unit – III	**Knowledge Representation and Reasoning**	**8 Hours**
Logical Agents, Propositional Logic, Propositional Theorem Proving, Effective Propositional Model Checking, Agents Based on Propositional Logic, First-Order Logic, Syntax and Semantics of First-Order Logic, Using First-Order Logic, Propositional vs First-Order Inference, Unification and Lifting, Forward Chaining, Backward Chaining, Resolution, Knowledge Representation, Reasoning Systems for Categories, Reasoning with Default Information, Truth Maintenance Systems, Closed World Assumption, Constraint Propagation and Backtracking Search		
Unit – IV	**Forms of Learning, Reinforcement Learning**	**8 Hours**
Learning from Examples, Forms of Learning, Supervised Learning, Learning Decision Trees, Evaluating and Choosing the Best Hypothesis, The Theory of Learning, Regression and Classification with Linear Models, Artificial Neural Networks, Nonparametric Models, Support Vector Machines, Ensemble Learning, Practical Machine Learning, Reinforcement Learning, Introduction, Passive Reinforcement Learning		

Internal Continuous Assessment (ICA):

ICA consists of a minimum of eight experiments performed on the above syllabus.

Text Books

1. Artificial Intelligence: A Modern Approach, Stuart Russell and Peter Norvig, 3rd Edition, Prentice Hall
2. A First Course in Artificial Intelligence, Deepak Khemani, McGraw Hill Education (India)
3. Introduction to Artificial Intelligence & Expert Systems, Dan W Patterson, PHI.

Reference Books

1. Artificial Intelligence, Elaine Rich and Kevin Knight, Tata McGraw Hill

23ETU7E22T: Program Elective IV - Architecting IoT Solutions

Teaching Scheme		**Examination Scheme**	
Lectures	2 Hours/week	**ESE**	60 Marks
Practical	2 Hours/week	**ISE**	40 Marks
Credits	3	**ICA**	25 Marks

Introduction:

This course introduces the AWS Well-Architected Framework and its role in designing efficient, secure, reliable, and high-performing cloud-based infrastructures. The course focuses on architectural best practices, workload design, performance considerations, and cost optimization using AWS services. Students gain practical exposure to evaluating and improving cloud architectures using the Well-Architected Framework.

Course Prerequisite:

This course requires knowledge of fundamentals of IoT and Cloud Platforms.

Course Objectives:

1. To introduce students to the AWS Well-Architected Framework and its architectural principles.
2. To understand the core pillars of the framework for building secure, reliable, and efficient cloud systems.
3. To explain security, reliability, and operational aspects involved in cloud architectures.
4. To analyze performance and cost considerations for multiple cloud workloads.

Course Outcomes:

At the end of the course, students will be able to

1. Apply AWS Well-Architected Framework principles to cloud architecture design.
2. Review and assess cloud architectures for operational, security, and reliability risks.
3. Monitor and evaluate performance efficiency for different cloud workloads.
4. Analyze cost and performance trade-offs in cloud infrastructure design.

Unit – I	**Introduction to AWS Well-Architected Framework**	**8 Hours**

Introduction and definitions, architecture concepts, general design principles, overview of AWS Well-Architected Framework, workload concepts and architectural best practices.

Unit – II	Core Pillars of the Well-Architected Framework	7 Hours
Design principles, definitions, best practices, and resources related to Operational Excellence, Security, Reliability, Performance Efficiency, and Cost Optimization.		
Unit – III	**Operations, Security, and Reliability**	**8 Hours**
Operational excellence concepts, organization, prepare, operate, and evolve phases, identity and access management, detection and monitoring, infrastructure protection, data protection, incident response, foundations of reliability, workload architecture, change management, and failure management.		
Unit – IV	**Performance Efficiency and Cost Optimization**	**7 Hours**
Performance efficiency concepts, resource selection, performance review and monitoring, trade-offs between compute and memory-intensive workloads, cloud financial management practices, expenditure and usage awareness, cost-effective resource utilization, demand and supply management, and continuous optimization.		

Internal Continuous Assessment (ICA):
ICA consists of a minimum of eight experiments performed on the above syllabus.

Text Books

1. AWS for Solutions Architects: Design your cloud infrastructure by implementing DevOps, containers, and Amazon Web Services, 2021, by Alberto Artasanchez, Packs Publishers 1st edition

Reference Books

1. AWS for System Administrators: Build, automate, and manage your infrastructure on the most popular cloud platform – AWS by Prashant Lakhera, Packt Publishing 1st edition

Additional Resources

1. AWS Well-Architected Framework Documentation
   https://docs.aws.amazon.com/wellarchitected/latest/framework/welcome.html

23ETU7E32T: Program Elective IV - Microwave Engineering

Teaching Scheme		Examination Scheme	
Lectures	2 Hours/week	**ESE**	60 Marks
Practical	2 Hours/week	**ISE**	40 Marks
Credits	3	**ICA**	25 Marks

Introduction:

This course introduces the importance of microwave engineering as emerging technology to be used for communication applications. It constitutes generation, transmission, and measurement of various parameters dealing with microwave frequency. The performance analysis is carried out using microwave network analysis.

Course Prerequisite:

This course requires knowledge of Electromagnetic Field Theory, Network Analysis, and basic Electronic Devices.

Course Objectives:

1. To make students aware about Microwave communication and transmission lines and it's parameters.
2. To identify different modes of microwave propagation for waveguide and wave parameters.
3. To list various methods of microwave generation and transmission using active and passive components.
4. To familiarize about measurement of various microwave parameters.

Course Outcomes:

At the end of the course, students will be able to

1. Describe microwave bands and hazards, and analyze transmission line behavior using R, L, C, G, Z_0, and γ to determine reflection, transmission, and standing wave characteristics.
2. Analyze different modes of propagation in waveguides and derive the associated wave parameters.
3. Analyze multiport microwave junctions and non-reciprocal devices using S-parameters and derive their scattering matrices.
4. Describe the working principles of all the microwave tubes and solid-state devices.
5. Measure microwave power, frequency, attenuation, and VSWR using appropriate microwave test instruments and techniques.

Unit – I	Introduction to Microwaves and Transmission Lines	6 Hours
Microwave frequency band, Characteristics and applications, Microwave hazards of microwaves. Transmission line sections as circuit elements, Transmission line equations using field theory and circuit theory, transmission line primary constants (R,L,C,G) and secondary constants (Z0, γ), Transmission line parameters (VSWR, Reflection coefficient, transmission coefficient)		
Unit – II	**Rectangular Waveguide**	**6 Hours**
Comparison between Transmission Line and Waveguide, TE/TM mode analysis, Waveguide parameters (fc, λc, λg, Vp, Vg), Relation between Vp and Vg, Power transmission in Rectangular Waveguide, Cavity resonators.		
Unit – III	**Microwave Passive Components**	**7 Hours**
Introduction to S parameters, Multi port junctions - Construction and operation of E-plane, H-plane, Magic Tee and Directional Coupler, S matrix for E-Plane Tee, H-Plane Tee, Magic Tee and Directional Coupler. Non reciprocal devices – Faraday's rotation - Construction and operation of Isolator and Circulator.		
Unit – IV	**Microwave Tubes & Solid State Devices**	**7 Hours**
Limitations of conventional circuits at microwave frequencies. Construction and principle of operation of Microwave Tubes: Two cavity Klystron, Reflex Klystron, Magnetron, TWT Principle of operation, specifications and applications of solid-state devices: Gunn Diode, IMPATT, TRAPATT diode.		
Unit – V	**Microwave Measurements**	**4 Hours**
Measurement of power, frequency, attenuation, VSWR.		
Internal Continuous Assessment (ICA): ICA consists of a minimum of eight experiments performed on the above syllabus.		
Text Books		
1. Microwave Devices and Circuits, Samuel Y. Liao, 3rd edition, Pearson 2. Microwave Engineering, David M. Pozar, Fourth edition, Wiley publications. 3. Microwave and Radar engineering, M. Kulkarni, 3rd edition, Umesh Publications		
Reference Books		
1. Foundations for Microwave Engineering by Robert Collin, Wiley publications 2. Microwave Engineering (Passive Circuit) by Peter Rizzi, Pearson Education 3. Basic Microwave Techniques and Laboratory Manual, M L Sisodia& G S Raghuvanshi, New age international (P) Limited, Publishers		

WALCHAND INSTITUTE OF TECHNOLOGY, SOLAPUR **(An Autonomous Institute)** **Final Year B.Tech. (Electronics and Telecommunication Engineering), Semester-VII**			
23ETU7E13T: Program Elective V - Business Intelligence			
Teaching Scheme		**Examination Scheme**	
Lectures	2 Hours/week	**ESE**	60 Marks
Tutorial	1 Hour/week	**ISE**	40 Marks
Credits	3	**ICA**	25 Marks

Introduction:

This course introduces basic components of Business Intelligence environments, discuss business analytics, data mining, data visualization, data tools and infrastructure and describe various applications of BI.

Course Prerequisite:

This course requires knowledge of basic types of data, data preprocessing methods, data cleaning and features extractions techniques.

Course Objectives:

1. To Introduce students with the basic components that make a business intelligence environment.
2. To Discuss the structure of the decision-making process
3. To explain the mathematical model for business intelligence.
4. To introduce different visualization tools and techniques for data representation and report preparation.
5. To discuss various applications of Business Intelligence

Course Outcomes:

At the end of the course, students will be able to

1. Describe the basic components of the BI environment.
2. Use ETL and BI tools for the decision support system.
3. Apply data mining techniques for data analysis.
4. Apply different visualization tools for report generation and explicate components of business performance measurement systems.
5. Illustrate various applications of Business Intelligence.

Unit – I	**Introduction to Business Intelligence**	**5 Hours**
Effective and timely decisions, role of mathematical models, BI architectures, ethics on BI, Introduction to data warehouse, architecture, Definition and architecture of Data warehouse, Cubes and multidimensional analysis, OLAP		

Unit – II	Decision Support System	5 Hours
Representation of decision-making system, evolution of information system, definition and development of decision support system, mathematical models for decision-making.		
Unit – III	**Data Mining**	**4 Hours**
Definition and applications of data mining, data mining process, analysis methodologies, data preparation validation, transformation, reduction and exploration.		
Unit – IV	**Business Reporting, Visual Analytics and Business Performance Management**	**6 Hours**
Business reporting definitions and concepts, data and information visualization, different types of charts and graphs, visual analytics, performance dashboards scorecards, business performance management		
Unit – V	**BI applications: Marketing Models Logistic Models**	**10 Hours**
Relational marketing, Salesforce management, Supply chain optimization, optimization models for logistics planning, revenue management system, case studies- Marketing models- Logistics business case studies		
Internal Continuous Assessment (ICA): ICA consists of minimum eight tutorials based upon above syllabus and a Mini Project		
Text Books		
1. Business Intelligence Data mining and optimization for Decision making by Carlo Vercellis, ISBN:978-81-265-4188-1, Wiley Publication. 2. Business Intelligence and Analytics: Systems for Decision Support by Efraim Turban, Ramesh Sharda, Dursun Delen by Pearson Education, Ltd 3. Data Mining and Business Intelligence by S.K. Shinde and Uddagiri Chandrashekhar 4. Data Mining for Business Intelligence by Galit Shmueli, Nitin Patel, Peter Bruce, Wiley Publications		
Reference Books		
1. Data Warehousing in the Real World – Anahory & Murray, Pearson Edt 2. Data Warehousing Fundamentals – Ponniah [Wiley Publication]		

23ETU7E23T: Program Elective V - Automotive Electronics

Teaching Scheme		**Examination Scheme**	
Lectures	2 Hours/week	**ESE**	60 Marks
Tutorial	1 Hour/week	**ISE**	40 Marks
Credits	3	**ICA**	25 Marks

Introduction:

Automotive Electronics plays a crucial role in modern vehicles by enabling intelligent control, safety, efficiency, and reliability. With the increasing use of electronic control units, sensors, embedded systems, and communication networks, vehicles have evolved into complex electronic systems on wheels. This course introduces Electronics and Telecommunication Engineering students to the fundamental electronic systems used in automobiles. The emphasis is on understanding core electronic components, sensing mechanisms, control units, in-vehicle communication, and basic electric vehicle electronics in a simple and self-learning-oriented manner.

Course Prerequisite:

This course requires basic knowledge of Analog and Digital Electronics, Microcontrollers and Sensors

Course Objectives:

1. To introduce basic concepts of automotive electronics and vehicle electronic systems.
2. To understand commonly used automotive sensors and actuators.
3. To explain the role of electronic control units and in-vehicle communication.
4. To introduce automotive safety electronics and electric vehicle basics.

Course Outcomes:

At the end of the course, students will be able to
1. Identify major electronic systems used in modern vehicles.
2. Explain the working of basic automotive sensors and actuators.
3. Describe ECU functions and basic in-vehicle communication concepts.
4. Understand safety electronics and basic electric vehicle electronics.

Unit – I	**Introduction to Automotive Electronics**	**8 Hours**

Overview of automotive electronics, evolution of vehicle electronic systems, classification of automotive electronic subsystems, role of electronics in engine, braking, steering, and body systems, basic automotive operating environment.

Unit – II	**Automotive Sensors and Actuators**	**7 Hours**
Common automotive sensors such as temperature, pressure, speed, and position sensors, basic working principles, signal conditioning basics, commonly used actuators including motors, solenoids, and relays, simple interfacing concepts.		
Unit – III	**Electronic Control Units and Vehicle Communication**	**7 Hours**
Introduction to electronic control units, basic ECU architecture, microcontrollers in automotive applications, introduction to in-vehicle communication, basics of Controller Area Network protocol, overview of LIN.		
Unit – IV	**Safety Electronics and Electric Vehicle Basics**	**8 Hours**
Basic automotive safety systems such as airbag and ABS electronics, introduction to onboard diagnostics, overview of electric and hybrid vehicles, basic electric vehicle architecture, introduction to battery management systems.		

Internal Continuous Assessment (ICA):

ICA consists of minimum eight tutorials based upon above syllabus

Text Books

1. William Ribbens, Understanding Automotive Electronics, Butterworth-Heinemann.
2. Tom Denton, Automobile Electrical and Electronic Systems, Routledge.

Reference Books

1. Robert Bosch GmbH, Automotive Handbook, Wiley.
2. James D. Halderman, Automotive Electricity and Electronics, Pearson Education.
3. Ronald K. Jurgen, Automotive Electronics Handbook, McGraw-Hill.

Additional Online Resources

1. Bosch Mobility Solutions – Technical Learning Resources
2. CAN Bus Documentation and Tutorials
3. NPTEL Online Courses on Automotive Electronics
4. Open Learning Resources on Electric Vehicle Systems

23ETU7E33T: Program Elective V - Mobile Communication

Teaching Scheme		**Examination Scheme**	
Lectures	2 Hours/week	**ESE**	60 Marks
Tutorial	1 Hour/week	**ISE**	40 Marks
Credits	3	**ICA**	25 Marks

Introduction:

This course provides comprehensive understanding of mobile communication, cellular concepts and practical applications in modern systems.

Course Prerequisite:

This course requires basic knowledge of telecommunications and networking concepts along with an understanding of digital signal processing and RF engineering fundamentals.

Course Objectives:

1. To introduce students' cellular communication principles.
2. To examine the Mobile radio propagation, cellular system design in mobile communication
3. To describe the components, hierarchical timing, capacity calculations, and service categories of the GSM Communications.
4. To describe an overview of 4G & 5G next generation technology.

Course Outcomes:

At the end of the course, students will be able to

1. Apply cellular communication principles to calculate frequency reuse factor and co-channel reuse distance.
2. Analyze various losses in mobile radio propagations and diffraction in mobile communication.
3. Describe GSM architecture. frame structure, system capacity and services provided by the GSM Communication.
4. Explicate emerging technologies required for fourth generation mobile systems such as Long Term Evolution (LTE) & 5G next generation technology.

Unit – I	**Introduction to Wireless Communication Systems**	**7 Hours**

The Cellular Engineering Fundamentals: Introduction, Frequency Re-use, Channel Assignment Strategies, Handoff Strategies, Interference and System Capacity, Co-channel Interference (CCI), Adjacent Channel Interference (ACI), Microcell Zone concept, Repeaters.

Unit – II	Mobile Radio Propagation	5 Hours
Large scale path loss, Free space propagation model, ground reflection model (two ray model), diffraction, Practical Link Budget using path loss scale multipath propagation.		
Unit – III	**GSM**	**5 Hours**
GSM Network architecture, signaling protocol architecture, identifiers, channels, Frame structure, speech coding, authentication and security, call procedure, handoff procedure, services and features. Mobile data networks, GPRS and higher data rates.		
Unit – IV	**CDMA digital cellular standard (IS-95) & IMT – 2020**	**6 Hours**
Frequency and channel specifications of IS-95, forward and reverse CDMA channel, packet and frame formats, mobility and radio resource management. IMT 2000 & IMT Advanced, IMT 2020, capabilities.		
Unit – V	**4G (LTE) & 5G Next Generation Technology**	**7 Hours**
Introduction to 4G, LTE Architecture, Elements of LTE- EPS, LTE Radio / air interface-Modulation and features, LTE Channels, Introduction to 5G, 5G CN Architecture, Radio/air interface, features		

Internal Continuous Assessment (ICA):
ICA consists of minimum eight assignment/tutorial based upon above syllabus.

Text Books

1. Wireless Communications - Theodore S. Rappaport, Prentice Hall of India, PTR Publication.
2. Principles of Wireless Networks – Kaveh Pahlavan, Prashant Krishnamurthy, PHI.
3. Mobile Communication – G. K. Behera & Lopamudra Das, Scitech Publication.
4. Mobile Communications – Jochen Schiller, Pearson Education, Second Edition.

Reference Books

1. Wireless Communication – Singhal, TMH.
2. Mobile and Personal Communication Systems and Services – Raj Pandya, Prentice Hall of India.
3. Wireless Communication – D. P. Agarwal, Thomnson learning 2007, Second Edition.
4. 4-G Roadmap and Emerging Communication Technologies – Young Kyun Kim and Ramjee Prasad –Artechhouse.
5. 5G NR: The Next Generation Wireless Access Technology- By Erik Dahlman, Stefan Parkvall, Johan Skold

23ETU7CC4T: Programmable Logic Control

Teaching Scheme		Examination Scheme	
Lectures	3 Hours/week	**ESE**	60 Marks
Practical	2 Hours/week	**ISE**	40 Marks
Credits	4	**ICA**	25 Marks

Introduction:

A programmable logic controller (PLC), also known as a programmable controller, is a digital computer used for automation of industrial electromechanical processes such as control of machinery on factory assembly lines, amusement rides, and lighting systems. It is also used in domestic applications like washing machines and elevator systems. The objective of this course is to introduce students to the functions, interfacing, and programming of PLCs, and the fundamentals of HMI and SCADA. This course also provides a comprehensive theoretical and applied foundation for industrial process control.

Course Prerequisite:

This course requires knowledge of Digital Devices, combinational and sequential logic circuit design and simulation.

Course Objectives:

1. To provide a comprehensive understanding of PLC fundamentals, system architecture, communication methods, and industrial applications.
2. To develop proficiency in PLC programming using ladder logic and control instructions for solving industrial automation problems.
3. To enable students to design and analyze PLC-based industrial control systems, including sensor–actuator interfacing and PLC communication and networking.
4. To impart knowledge of process control principles, controllers, sensors, actuators, signal conditioning, and HMI–SCADA based industrial automation systems.

Course Outcomes:

After completing this course, student will be able to -

1. Explain the architecture, components, communication techniques, and industrial applications of Programmable Logic Controllers (PLCs).
2. Develop, test, and debug PLC ladder logic programs using logic functions, timers, counters, and control instructions.
3. Design PLC-based industrial automation systems incorporating sensor–actuator interfacing and basic PLC communication and networking.
4. Analyze industrial process control systems and apply appropriate controllers, sensors, actuators, signal conditioning techniques, and HMI–SCADA concepts.

Unit – I	Introduction to PLC and System Architecture	07 Hours

Controllers, microprocessor-based controllers, PLC overview, applications, typical PLC system, internal architecture, memory organization, I/O modules, ladder diagrams and symbols.

Unit – II	**PLC Programming and Logic Functions**	**07 Hours**
PLC programming concepts, ladder logic, logic functions, latching, interlocking, internal relays, timers, counters, jump and call instructions, programming examples.		
Unit – III	**Advanced PLC Programming and Industrial Interfacing**	**07 Hours**
Loop instructions, shift registers, data manipulation, PLC system design methodology, temperature control, valve sequencing, conveyor belt control, interfacing of sensors and actuators.		
Unit – IV	**Process Control, Sensors, Actuators & Signal Conditioning**	**07 Hours**
Fundamentals of process control, on–off, P, PI, PD, PID controllers, tuning methods, industrial sensors, actuators, hydraulic and pneumatic systems, signal conditioning circuits, data acquisition systems.		
Unit – V	**PLC Communication & Networking**	**07 Hours**
Need for communication in industrial automation systems, PLC communication architecture, Communication between PLCs, controllers, and remote I/O modules. Industrial communication protocols: Modbus RTU and Modbus TCP/IP. PLC system networking concepts. Serial and Ethernet-based PLC communication.		
Unit – VI	**HMI, SCADA & Industrial Automation Applications**	**07 Hours**
Introduction to HMI and SCADA systems SCADA architecture and components PLC–HMI communication SCADA configuration basics Industrial automation applications using PLC–SCADA		

Internal Continuous Assessment (ICA):

ICA consists of a minimum of eight experiments based on the above syllabus.

Text Books

1. Programmable Logic Controllers; Bolton, Elsevier-Newnes, 3rd Edition
2. Programmable Logic Controllers Programming Methods and Applications; John R. Hack Worth, Frederick D. Hackworth, Jr.; Prentice Hall India
3. Industrial & Process Control, C.D. Johnson, John Wiley & Sons Inc, Eight Edition
4. Industrial Electronics: Circuits, instruments and control techniques, Terry Bartelt, Delmar Publishers

Reference Books

1. Programmable Logic Controllers and applications; John W Webb Ronald A. Reis, PHI Learning
2. Programmable Logic Controllers, Frank Petruzella, McGraw-Hill Higher Education
3. Programmable Logic Controllers Gray Durming, Third Edition

WALCHAND INSTITUTE OF TECHNOLOGY, SOLAPUR
(An Autonomous Institute)

Final Year B.Tech. (Electronics and Telecommunication Engineering), Semester-VII

23ETU7PR5L: Project Phase I

Teaching Scheme		**Examination Scheme**	
Lectures		**ESE**	-
Practical	8 Hours/week	**ICA**	100 Marks
Credits	4	**POE**	50 Marks

Introduction:

Project-based learning is a well-established educational paradigm gaining increasing importance today. To align with this approach, a project course is integrated into the final year curriculum, spanning both semesters. In this course, students work in teams to undertake a project, allowing them to showcase their abilities and develop expertise in their chosen areas of interest. The projects can involve both hardware and software, with a strong emphasis on design and research aspects. Additionally, effective verbal and written communication is a crucial skill for engineering graduates in various contexts. This course aims to cultivate these essential communication skills as well.

Course Prerequisite:

A student must possess both technical competency and effective teamwork skills to successfully contribute to a project. This includes adept knowledge of hardware and software architecture, as well as associated programming skills. Additionally, the student should have strong technical report writing and presentation abilities.

Course Objectives:

1. To expose students to the electronics and software engineering industries, enabling them to identify problem areas and formulate problem statements.
2. To equip students with the ability to design electronic and software systems that address the identified problems
3. To develop students' skills in designing the hardware and software architecture for their projects.
4. To train students in writing technical specifications and project documentation for their chosen problems

Course Outcomes:

After completing the course, students will be able to

1. Conduct a comprehensive literature review to identify a project that addresses societal and environmental needs for sustainable development.
2. Develop a detailed project plan, including a timeline and an estimated budget.
3. Utilize engineering expertise to design the implementation architecture required for the project.
4. Exhibit teamwork and presentation skills by preparing thorough reports and delivering presentations, while adhering to professional ethical standards.

Guidelines:

1. The student will finalize the project after obtaining approval from their guide and submit a synopsis along with a presentation.
2. The student should prepare the project design.
3. The project synopsis should ideally include an abstract, literature survey, problem definition, and proposed system and design.
4. The student will need to present his/her work on the project design implemented and submit project phase -I report with details of implementation methodology, results and discussion.

WALCHAND INSTITUTE OF TECHNOLOGY, SOLAPUR (An Autonomous Institute) Final Year B.Tech. (Electronics and Telecommunication Engineering), Semester-VIII			
23CMU8RM1T: Research Methodology and Intellectual Property Rights			
Teaching Scheme		**Examination Scheme**	
Lectures	2 Hours/week	**ESE**	50 Marks
Credits	2		

Introduction:

Research Methodology provides a systematic framework for identifying, analyzing, and solving technical problems through evidence-based inquiry. Intellectual Property Rights (IPR) establish the legal safeguards necessary to protect these engineering innovations, ensuring that creators maintain control and commercial benefits over their inventions. Together, they equip students with the tools to conduct ethical research and navigate the global patent and copyright landscape.

Course Prerequisite:

Basics of Engineering and Statistics

Course Objectives:

1. Impart knowledge on research types, processes, design, and literature review techniques.
2. Develop skills in data collection, sampling, and statistical analysis of research data.
3. Enable preparation of research reports with standard citations and inculcate research ethics.
4. Provide understanding of IPR types, legal frameworks, patent search, and filing process.

Course Outcomes:

At the end of the course, students will be able to

1. Explain the fundamental concepts of research design, its types, processes, and systematic literature review techniques.
2. Analyze research data by using appropriate data collection methods, sampling techniques, and statistical tools.
3. Prepare research reports using standard citation formats and demonstrate ethical responsibilities in research conduct.
4. Classify IPR types and legal frameworks, and apply knowledge to patent search and filing process.

Unit – I	**Introduction to Research**	**4 Hours**
Meaning & objectives of research, Types of research: basic, applied, quantitative, qualitative, Steps in research process, Identifying research problems, Formulating hypotheses and objectives		
Unit – II	**Literature Review & Research Design**	**5 Hours**
Literature search strategies, IEEE Xplore, Scopus, Google Scholar, Springer, Web of Science; Research gap analysis, Research design: exploratory, descriptive, experimental, Sampling techniques		
Unit – III	**Data Collection & Analysis**	**5 Hours**
Primary vs secondary data, Tools of data collection (questionnaires, surveys, interviews), Basics of statistical analysis: Mean, Median, Mode, Standard deviation, correlation; Use of software (Excel, SPSS), Data interpretation and presentation.		
Unit – IV	**Technical Communication & Ethics in Research**	**4 Hours**
Technical writing framework, Research paper structure: Abstract, Introduction, Methodology, Results, Discussion, Conclusion; Referencing and citations (APA/IEEE formats), Ethical considerations & plagiarism, Use of plagiarism checking tools		
Unit – V	**Introduction to Intellectual Property Rights**	**5 Hours**
IPR definitions & relevance to engineering, Patents: types, patentable subject matter; Copyright: scope, ownership, duration; Trademarks, Trade Secrets, Industrial Designs		
Unit – VI	**Patent Rules, Filing, and IP Management**	**5 Hours**
Patent search strategies, Patent drafting basics, Patent filing procedure in India and abroad, Patent Cooperation Treaty (PCT) overview, IPR protection strategy for products and software, Technology transfer, Licensing		

Text Books

1. Research Methodology, A Practical and Scientific Approach - Vinayak Bairagi, Mousami V. Munot, Chapman and Hall
2. Research Methodology: C.R. Kothari , New Age International Publications
3. Research methodology: A step-by-step guide for beginners - Kumar, R. (2018)
4. Intellectual Property Rights: P. Narayanan, S. Chand Publication

Reference Books

1. “Research Methodology: Methods & Techniques” — Ranjit Kumar, Sage Publications
2. Intellectual Property Rights under WTO, T. Ramappa
3. Intellectual Property Handbook, WIPO (World Intellectual Property Organization, free PDF). : https://tind.wipo.int/record/28661/files/wipo_pub_489.pdf

Relevant Videos on Youtube related to previous NPTEL Courses
1) Research Methodology — NPTEL course (Videos Available on Toutube) https://www.youtube.com/watch?v=E2gGF1rburw&list=PLyqSpQzTE6M8F_P8lgjvmqiDEoFGLzG4h&index=1   2) Intellectual Property – NPTEL course (Videos Available on Toutube) https://www.youtube.com/watch?v=a-zBaQ7Al2s&list=PLyqSpQzTE6M8PuzP1p2hNPXgpbOBhFgja
Important Websites / E-Resources
1) IEEE Xplore Digital Library: https://india.ieee.org/ieee-xplore-digital-library/   2) Google Scholar: https://scholar.google.com/
Patent Databases (url)
1) WIPO (World Int'l Patents): https://www.wipo.int/en/web/patents   2) Indian Patent Advanced Search System (InPASS): https://iprarch.ipindia.gov.in/publicsearch

23ETU8E12T: Program Elective VI - Artificial Intelligence Applications

Teaching Scheme		Examination Scheme	
Lectures	2 Hours/week	**ESE**	50 Marks
Credits	2		

Introduction:

This course provides an introduction to the fundamental concepts and applications of Artificial Intelligence. It covers key areas such as intelligent systems, expert systems, robotics, and Generative AI, with emphasis on real-world applications and ethical considerations. The course enables students to understand how AI technologies are applied to solve complex problems across industry and society

Course Prerequisite:

This course requires knowledge of Artificial Intelligence and Machine Learning, Elementary understanding of probability and logic.

Course Objectives:

5. To introduce the fundamental concepts, scope, and application areas of Artificial Intelligence.
6. To familiarize students with the principles, structure, and applications of expert systems.
7. To develop understanding of robotic systems, including perception, motion planning, and control.
8. To provide insights into Generative AI technologies along with their applications and ethical considerations.

Course Outcomes:

At the end of the course, students will be able to

5. Demonstrate understanding of core concepts and applications of Artificial Intelligence.
6. Describe the structure, knowledge representation, and applications of expert systems.
7. Analyze components, perception, and motion planning in robotic systems.
8. Explore applications and ethical issues of Generative AI in society and industry.

Unit – I	**Overview of AI Applications**	**6 Hours**
Introduction, definitions of AI, Man Vs Computers, Simulation of Sophisticated and Intelligent Behavior, Branches of AI, Natural Language, Automated reasoning, Visual perception.		
Unit – II	**Expert Systems and Applications**	**8 Hours**
Introduction to Expert system, Knowledge Representation, Expert System Shells, Knowledge Acquisition of an Expert System, Applications of Expert Systems, Examples of Expert Systems		
Unit – III	**Robotics**	**8 Hours**
Introduction, Robot Hardware, Robotic Perception, Planning to Move, Planning Uncertain Movements, Moving, Robotic Software Architectures, Application Domains.		

<table>
<tr><td>Unit – VI</td><td>Generative AI and Applications</td><td>6 Hours</td></tr>
<tr><td colspan="3">Definition and Importance of Generative AI, Types of Generative Models. Applications Art and Design: Generative Art, Music Generation, Text Generation: Chatbots, Story Generation, Image and Video Synthesis: Deepfakes, Style Transfer, Research and Industry Applications
Ethical Issues: Deepfakes, Copyright, Misinformation, Bias and Fairness in AI, AI and Privacy, Responsible Use and Regulation, Future of AI and Society</td></tr>
<tr><td colspan="3">Text Books</td></tr>
<tr><td colspan="3">1. “A Classical Approach to Artificial Intelligence” by Munesh Chandra Trivedi, Second Edition, Khanna Publishing.
2. "Artificial Intelligence: A Modern Approach" by Stuart Russell and Peter Norvig (Third edition) Pearson publication
3. "Artificial Intelligence Concepts and Applications by Lavika Goel, Wiley Publications.</td></tr>
<tr><td colspan="3">Reference Books</td></tr>
<tr><td colspan="3">1. Artificial Intelligence by Saroj Koushik, Cengage Learning Publication
2. The Essence of Artificial Intelligence by Alison Cousey, Pearson Publication
3. Introduction to Artificial Intelligence and Expert systems by Dan W. Patterson, Pearson</td></tr>
</table>

23ETU8E22T: Program Elective VI - Industrial Internet of Things

Teaching Scheme		Examination Scheme	
Lectures	2 Hours/week	**ESE**	50 Marks
Credits	2		

Introduction:

Industrial Internet of Things (IIoT) focuses on the application of IoT concepts in industrial environments involving sensors, control systems, communication networks, and data-driven decision making. This course introduces students to the fundamentals of IIoT architecture, industrial communication technologies, and security considerations. The emphasis is on conceptual understanding of industrial systems rather than platform-specific implementations, making it suitable for self-learning by Electronics and Telecommunication Engineering students.

Course Prerequisite:

This course requires knowledge of communication systems.

Course Objectives:

1. To introduce the concept of Industrial IoT and its distinction from consumer IoT.
2. To explain the architecture and core components of Industrial IoT systems.
3. To understand industrial communication technologies and associated security concerns.
4. To provide an overview of cloud integration concepts used in Industrial IoT systems

Course Outcomes:

At the end of the course, students will be able to

1. Explain the fundamental concepts and components of an Industrial IoT system.
2. Describe standard reference architectures used in Industrial IoT.
3. Identify appropriate communication technologies for industrial applications.
4. Analyze basic security challenges in Industrial IoT environments.

Unit – I	Introduction to Industrial IoT	7 Hours

Concept of Industrial IoT, evolution from IoT to IIoT, differences between IoT and IIoT, key drivers and benefits of IIoT, technical requirements of industrial systems, overview of industrial sensors and data sources, role of IIoT in modern industries.

Unit – II	Industrial IoT Reference Architecture	8 Hours
Industrial Internet Reference Architecture, Industrial Internet Architecture Framework, architectural layers and components, three-tier architecture, connectivity concepts, data acquisition and data management, role of gateways in IIoT systems, interaction between operational technology and information technology.		
Unit – III	**Industrial Communication Technologies and Security**	**8 Hours**
Overview of industrial communication requirements, low-power WAN technologies for IIoT such as LoRaWAN, SigFox, LTE Category-M, comparison of wired and wireless industrial communication, introduction to industrial protocols, security challenges in industrial environments, network-level and system-level security issues, identity and access management concepts.		
Unit – IV	**Cloud Integration Concepts for Industrial IoT**	**10 Hours**
Role of cloud platforms in Industrial IoT, general cloud-based IIoT architecture, data ingestion and storage concepts, analytics overview, comparison of industrial cloud platforms, limitations and challenges of cloud-based IIoT systems, overview of open and proprietary cloud solutions.		

Text Books

1. Industry 4.0: The Industrial Internet of Things, Alasdair Gilchrist, Apress, 1st Edition.
2. Hands-On Industrial Internet of Things, Giacomo Veneri, Antonio Capasso, Packt Publishing, 1st Edition.
3. Internet of Things: A Hands-On Approach, Arshdeep Bahga, Vijay Madisetti, Orient Blackswan, 1st Edition

Reference Books

1. Rethinking the Internet of Things: A Scalable Approach to Connecting Everything, Francis daCosta, Apress Publications, 1st Edition

Additional Online Resources

1. AWS IoT Core Documentation
2. OpenStack Documentation
3. Azure IoT Documentation

23ETU8E32T: Program Elective VI - Programmable ICs and ASIC

Teaching Scheme		Examination Scheme	
Lectures	2 Hours/week	**ESE**	50 Marks
Credits	2		

Introduction:

This course provides a deep understanding of programmable ASICs, FPGA-based systems, SoC design principles and Complex Programmable Logic Device (CPLD), equipping with valuable skills for advanced electronic engineering applications.

Course Prerequisite:

This course requires knowledge of Digital Devices, CMOS logic.

Course Objectives:

1. To introduce learn concept of ASIC, Programable ICs and its library cell design.
2. To analyze Programmable ASIC Logic and I/O Cells for design of various memory devices.
3. To make students understand architecture of FPGA and CPLDs for design of digital systems.
4. To explain the concept of SoC and explore its applications.

Course Outcomes:

At the end of the course, students will be able to

1. Describe ASIC and programmable IC types, cell design flow and interconnects.
2. Implement Boolean and digital logic functions using different Programmable Logic Devices.
3. Comprehend the architecture of FPGA for implementation of digital system design.
4. Explore the applications of SoC in specific domains.

Unit – I	Introduction to ASIC and Programmable ICs	8 Hours

Types of ASICs & design flow, ASIC library cell design: standard cell design, Gate array design, Antifuse, SRAM based ASICs, Programmable ASIC logic cells and I/O cells, Programmable interconnects.

Unit – II	Programmable Logic Devices and CPLDs	8 Hours
	EPROM, PLA, PAL– architecture & Features, Implementation of Boolean functions using PLDs, CPLD architecture, Comparison of PLDs, CPLDs, and FPGAs, Commercial CPLD Devices: Xilinx XC9500, Altera Max7000.	
Unit – III	**FPGA based system Design**	**8 Hours**
	Introduction to FPGA-based digital design, FPGA organization and architecture, FPGA fabrics – SRAM-based and non-volatile FPGAs, Logic blocks, routing resources, and I/O blocks, Digital design flow using FPGA, Commercial FPGA devices: Altera Flex 10k, Actel ACT -Xilinx LCA.	
Unit – IV	**System-on-Chip (SOC) Design**	**6 Hours**
	Voice over IP SOC, Intellectual Property, SOC Design challenges- Methodology and design-FPGA to ASIC conversion, SOC design-verification-Testing, Set top box SOC.	

Text Books

1. Application Specific Integrated Circuits, by M. J. S. Smith, Pearson Education.
2. FPGA-Based System Design, by Wayne Wolf, Prentice Hall PTR.
3. Digital Design Using Field Programmable Gate Arrays, by Pak K. Chan & S. Mourad, PTR Prentice Hall .

Reference Books

1. Digital Integrated Circuits, Rabey, Chandrakasan, Nikolic, Pearson Education.
2. From ASICs to SOCs: A Practical Approach, by Farzad Nekoogar and Faranak Nekoogar, Prentice Hall PTR.
3. Principles of CMOS VLSI Design, Neil Weste, Kamran Eshraghian, Addison Wesley/Pearson Education.
4. Modern VLSI Design, Wayne Wolf , Kamran Eshraghian, Prentice Hall,.
5. Essentials of VLSI Circuits and Systems, Ehraghian, Dauglas A. Pucknell and Sholeh Eshraghiam, – PHI, EEE.

23ETU8IN3L: Internship

Teaching Scheme		**Examination Scheme**	
Practical	20 Hours/week	**ICA**	100 Marks
Credits	10	**OE**	50 Marks

Introduction:

Internships serve as crucial educational and career development experiences, offering hands-on learning in specific fields or disciplines. They play a significant role in equipping individuals with essential industry skills, awareness of professional practices, and familiarity with organizational culture. Typically, structured and short-term, internships / On Job Training provide supervised training centered on particular tasks or projects, adhering to defined timelines.

Course Prerequisite:

This requires knowledge of core and multidisciplinary concepts, basic knowledge of tools and technologies, problem solving and analytical thinking skills, communication skills, teamwork and collaboration skills.

Course Objectives:

1. To provide students with hands-on experience in industrial environments, essential for developing industry-ready professionals.
2. To equip students with real-time technical and managerial skills necessary for their future careers.
3. To familiarize students with various materials, processes, products, and software applications, emphasizing quality control principles.
4. To introduce students to engineering ethics and responsibilities in professional practice.
5. To explore the social, economic, and administrative factors that impact industrial organizations' working environment.

Course Outcomes:

After completing the course, students will be able to

1. Develop professional competence through an internship experience.
2. Apply academic knowledge effectively in both personal and professional environment.
3. Expand their professional network and gain exposure to potential future employers.
4. Demonstrate the application of professional and societal ethics in their daily lives.
5. Explore their career goals and align them with personal aspirations

Guidelines:

Internship

Engineering internships are intended to provide students with an opportunity to apply theoretical knowledge from academics to the realities of the field work/training. The following guidelines are proposed to give academic credit for the internship undergone as a part of the Final Year Engineering curriculum.

1. Students may undergo internship related to engineering with Small/ Medium / Large scale industries to make themselves ready for the industry
2. Students may complete an Internship of a minimum of two months duration at the industry during Final Year Sem VIII.
3. The industry shall appoint a Supervisor to assess the performance of the student and share the same with the departmental supervisor for the fulfilment of ICA marks
4. Interns are required to have bi-weekly communication with their internal guide.
5. During the internship, the student will present a seminar in online /offline mode based on training in the presence of an expert committee from the department.
6. The student shall prepare a report of the work completed at the industry duly endorsed by the industry supervisor and submit the same as an internship report.
7. The Oral Examination for Internship shall be conducted by the departmental supervisor in the presence of an external industry or academic expert.

WALCHAND INSTITUTE OF TECHNOLOGY, SOLAPUR (An Autonomous Institute) Final Year B.Tech. (Electronics and Telecommunication Engineering), Semester-VIII			
23ETU8PR4L: Project Phase II			
Teaching Scheme		**Examination Scheme**	
Lectures		**ESE**	-
Practical	4 Hours/week	**ICA**	100 Marks
Credits	2	**POE**	50 Marks

Introduction:

Project-based learning is a well-established educational paradigm gaining increasing importance today. To align with this approach, a project course is integrated into the final year curriculum, spanning both semesters. In this course, students work in teams to undertake a project, allowing them to showcase their abilities and develop expertise in their chosen areas of interest. The projects can involve both hardware and software, with a strong emphasis on design and research aspects. Additionally, effective verbal and written communication is a crucial skill for engineering graduates in various contexts. This course aims to cultivate these essential communication skills as well.

Course Prerequisite:

A student must possess both technical competency and effective teamwork skills to successfully contribute to a project. This includes adept knowledge of hardware and software architecture, as well as associated programming skills. Additionally, the student should have strong technical report writing and presentation abilities.

Course Objectives:

1. To make student apply design concept, prepare detailed planning to solve problem undertaken
2. To make student to evaluate and analyze performance of the proposed solution to the problem undertaken
3. To make student aware of his responsibilities working in a team to provide time bound solutions to the problem
4. To make student write technical specifications, project document over problem undertaken.
5. To make student demonstrate a sound technical presentation of their selected project topic.
6. To make student aware of different software tools and soft-skills required to practice at various stages of project execution.

Course Outcomes:
After completing the course, students will be able to 1. Apply different design concepts to plan solution to the problem undertaken 2. Evaluate performance and detailed analysis of outcome of the proposed solution for the problem undertaken 3. Work in project group following work ethics 4. Communicate with engineers and the community at large 5. Demonstrate the knowledge, skills and attitudes of a professional engineer. 6. Select and use proper programming solution, simulator and necessary soft skills to provide solution to problem undertaken.
Guidelines: The objective of Project- II is to enable the student to extend further the investigative study taken up under Project-I, either fully practical or involving both theoretical and practical work, under the guidance of a supervisor from the department alone or jointly with a Supervisor drawn from R&D laboratory/Industry. This is expected to provide a good training for the student(s) in R&D work and technical leadership. The assignment normally includes: 1. In depth study of the topic assigned in the light of the report prepared under project-I 2. Review and finalization of the approach to the problem relating to the assigned topic 3. Detailed analysis/modelling/simulation/design/problem solving/experiment as needed 4. Final development of product/process, testing, results, conclusions and future directions 5. Preparing a paper for conference presentation/publication in journals and for project competition, if possible. 6. Preparing a project document in the standard format for being evaluated by the department. 7. Final Oral presentation and demonstration of project before a departmental and evaluation committee